



SIMATS Online Programming Lab
C PROGRAMMING

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QUESTIONS 25 / 25 completed
Remove
REMOVE
Write a C program to remove an element from an array.

PROGRAM EDITOR
C
Run Save

```

1 #include<stdio.h>
2 int main(){
3     int a[10],n,x,i;
4     scanf("%d",&n);
5     for(i=0;i<n;i++)scanf("%d",&a[i]);
6     scanf("%d",&x);
7     for(i=0;i<n;i++)
8         if(a[i]==x)
9             break;
10    for(i=i<n-1;i++)
11        a[i]=a[i+1];
12    for(i=0;i<n-1;i++)
13        printf("%d ",a[i]);
14 }
```

INPUT
5
1 2 3 4 5
3



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QUESTIONS 25 / 25 completed
Insert in Sorted
INSERT IN SORTED
Write a C program to insert an element in a sorted array.

PROGRAM EDITOR
C
Run Save

```

1 #include<stdio.h>
2 int main(){
3     int a[10],n,x,i;
4     scanf("%d",&n);
5     for(i=0;i<n;i++)scanf("%d",&a[i]);
6     scanf("%d",&x);
7     for(i=n-1;i>=0&& a[i]>x;i--)
8         a[i+1]=a[i];
9     a[i+1]=x;
10    for(i=0;i<=n;i++)
11        printf("%d ",a[i]);
12 }
```

INPUT
5
10 20 30 40 50
43



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QUESTIONS 25 / 25 completed
RM Duplicate
RM DUPLICATE
Write a C program to remove all the occurrences of an element from an array.

PROGRAM EDITOR
C
Run Save

```

1 #include<stdio.h>
2 int main(){
3     int a[10],n,x,i,j;
4     scanf("%d",&n);
5     for(i=0;i<n;i++)scanf("%d",&a[i]);
6     scanf("%d",&x);
7     for(i=0;i<n;i++)
8         if(a[i]==x){
9             for(j=i;j<n-1;j++)a[j]=a[j+1];
10            n--; i--;
11        }
12    for(i=0;i<n;i++)printf("%d ",a[i]);
13 }
```

INPUT
5
1 2 3 4 2
2



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QUESTIONS 25 / 25 completed

Palindrome

PALINDROME

Write a C program to check if an array is a palindrome or not.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main(){
3     int a[10],n,i,f=1;
4     scanf("%d",&n);
5     for(i=0;i<n;i++)scanf("%d",&a[i]);
6     for(i=0;i<n/2;i++)
7     if(a[i]!=a[n-1-i])f=0;
8     if(f)printf("Palindrome");
9     else printf("Not Palindrome");
10 }
```

INPUT

5
1 3 4 3 1



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QUESTIONS 25 / 25 completed

Common

COMMON

Write a C program to find the common elements between two arrays.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main(){
3     int a[10],b[10],n,m,i,j;
4     scanf("%d%d",&n,&m);
5     for(i=0;i<n;i++)scanf("%d",&a[i]);
6     for(i=0;i<m;i++)scanf("%d",&b[i]);
7     for(i=0;i<n;i++)
8     for(j=0;j<m;j++)
9     if(a[i]==b[j])printf("%d ",a[i]);
10 }
```

INPUT

5 5
1 2 3 4 5
4 6 2 7 1



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QUESTIONS 25 / 25 completed

Add 2 Matrix

ADD 2 MATRIX

Write a C program to add two matrices.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[3][3]={3,6,7},{4,6,9},{2,5,8}};
5     int b[3][3]={2,5,-9},{5,4,6},{8,6,7}};
6     int c[3][3],i,j;
7     for(i=0;i<3;i++)
8     {
9         for(j=0;j<3;j++)
10        {
11            c[i][j]=a[i][j]+b[i][j];
12            printf("%d",c[i][j]);
13        }
14        printf("\n");
15    }
16    return 0;
17 }
```

INPUT

Your INPUT goes here!



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QUESTIONS 25 / 25 completed

Multiply Matrix

MULTIPLY MATRIX

Write a C program to multiply two matrices.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[3][3]={3,6,7},{4,6,9},{2,5,8}};
5     int b[3][3]={2,5,-9},{5,4,6},{8,6,7}};
6     int c[3][3],i,j;
7     for(i=0;i<3;i++)
8     {
9         for(j=0;j<3;j++)
10        {
11            c[i][j]=0;
12            for(int k=0;k<3;k++)
13                c[i][j]+=a[i][k]*b[k][j];
14            printf("%d",c[i][j]);
15        }
16        printf("\n");
17    }
18    return 0;
19 }
```

INPUT

Your INPUT goes here!



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QUESTIONS 25 / 25 completed

Transpose

TRANPOSE

Write a C program to find the transpose of a matrix.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main(){
3     int a[3][3],i,j;
4     for(i=0;i<3;i++)
5     for(j=0;j<3;j++)
6     scanf("%d",&a[i][j]);
7     for(i=0;i<3;i++){
8     for(j=0;j<3;j++)
9     printf("%d ",a[j][i]);
10    printf("\n");
11 }
12 }
```

INPUT

1 2 3
4 5 6
7 8 9



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QUESTIONS 25 / 25 completed

Sum Diagonal

SUM DIAGONAL

Write a C program to find the sum of the diagonal elements of a matrix.

A matrix is a two-dimensional array of numbers arranged in rows and columns. The diagonal elements of a matrix are the elements that lie on the main diagonal, which starts from the top-left corner and goes down to the bottom-right corner.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[3][3]={3,6,7},{4,6,9},{2,5,8}};
5     int b[3][3]={2,5,-9},{5,4,6},{8,6,7}};
6     int c[3][3],i,j;
7     for(i=0;i<3;i++)
8     {
9         for(j=0;j<3;j++)
10        {
11            c[i][j]=a[i][j]+b[i][j];
12            printf("%d",c[i][j]);
13        }
14        printf("\n");
15    }
16    return 0;
17 }
```

INPUT

Your INPUT goes here!



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QUESTIONS 25 / 25 completed

Product Diag. ▾

PRODUCT DIAG.

Write a C program to find the product of the diagonal elements of a matrix.

PROGRAM EDITOR

C ▾

Run Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[3][3]={{{3,6,7},{4,6,9},{2,5,8}}};
5     int i,product=1;
6     for(i=1;i<3;i++)
7         product*=a[i][i];
8     printf("product of digaonal=%d",product);
9     return 0;
10 }
```

INPUT

Your INPUT goes here!



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QUESTIONS 25 / 25 completed

Sum Up Tri. ▾

SUM UP TRI.

Write a C program to find the sum of the upper triangular elements of a matrix.

PROGRAM EDITOR

C ▾

Run Save

```
1 #include<stdio.h>
2 int main(){
3     int a[3][3],i,j,s=0;
4     for(i=0;i<3;i++)
5         for(j=0;j<3;j++)
6             scanf("%d",&a[i][j]);
7     for(i=0;i<3;i++)
8         for(j=i;j<3;j++)
9             s+=a[i][j];
10    printf("%d",s);
11 }
```

INPUT

1 2 3
4 5 6
7 8 9



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QUESTIONS 25 / 25 completed

Sum Low Tri. ▾

SUM LOW TRI.

Write a C program to find the sum of the lower triangular elements of a matrix.

PROGRAM EDITOR

C ▾

Run Save

```
1 #include<stdio.h>
2 int main(){
3     int a[3][3],i,j,s=0;
4     for(i=0;i<3;i++)
5         for(j=0;j<3;j++)
6             scanf("%d",&a[i][j]);
7     for(i=0;i<3;i++)
8         for(j=0;j<=i;j++)
9             s+=a[i][j];
10    printf("%d",s);
11 }
```

INPUT

1 2 3
4 5 6
7 8 9



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QUESTIONS 25 / 25 completed
Sum Left Diag.
SUM LEFT DIAG.
Write a C program to find the sum of the elements in the left diagonal of a matrix.

PROGRAM EDITOR

C

Run

Save

```

1 #include<stdio.h>
2 int main(){
3     int a[3][3],i,j,s=0;
4     for(i=0;i<3;i++)
5         for(j=0;j<3;j++)
6         scanf("%d",&a[i][j]);
7     for(i=0;i<3;i++)
8         s+=a[i][i];
9     printf("%d",s);
10 }

```

INPUT

```

1 2 3
4 5 6
7 8 9

```



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QUESTIONS 25 / 25 completed
Sum Right Diag.
SUM RIGHT DIAG.
Write a C program to find the sum of the elements in the right diagonal of a matrix.

PROGRAM EDITOR

C

Run

Save

```

1 #include<stdio.h>
2 int main(){
3     int a[3][3],i,j,s=0;
4     for(i=0;i<3;i++)
5         for(j=0;j<3;j++)
6         scanf("%d",&a[i][j]);
7     for(i=0;i<3;i++)
8         s+=a[i][2-i];
9     printf("%d",s);
10 }

```

INPUT

```

1 2 3
4 5 6
7 8 9

```



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QUESTIONS 25 / 25 completed
Symmetric
SYMMETRIC
Write a C program to check if a matrix is symmetric or not.

PROGRAM EDITOR

C

Run

Save

```

1 #include<stdio.h>
2 int main(){
3     int a[3][3],i,j,f=1;
4     for(i=0;i<3;i++)
5         for(j=0;j<3;j++)
6         scanf("%d",&a[i][j]);
7     for(i=0;i<3;i++)
8         for(j=0;j<3;j++)
9         if(a[i][j]!=a[j][i]) f=0;
10    if(f) printf("Symmetric");
11    else printf("Not Symmetric");
12 }

```

INPUT

```

1 2 3
2 6 5
3 5 7

```



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QUESTIONS 25 / 25 completed

Diagonal or Not

DIAGONAL OR NOT

Write a C program to check if a matrix is diagonal or not.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main(){
3     int a[3][3],i,j,f=1;
4     for(i=0;i<3;i++)
5         for(j=0;j<3;j++)
6             scanf("%d",&a[i][j]);
7     for(i=0;i<3;i++)
8         for(j=0;j<3;j++)
9             if(i!=j&&a[i][j]!=0) f=0;
10    if(f) printf("Diagonal");
11    else printf("Not Diagonal");
12 }
```

INPUT

```
1 0 0
0 2 0
0 0 1
```



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QUESTIONS 25 / 25 completed

Up Tri. Not

UP TRI. NOT

Write a C program to check if a matrix is upper triangular or not.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main(){
3     int a[3][3],i,j,f=1;
4     for(i=0;i<3;i++)
5         for(j=0;j<3;j++)
6             scanf("%d",&a[i][j]);
7     for(i=1;i<3;i++)
8         for(j=0;j<i;j++)
9             if(a[i][j]!=0) f=0;
10    if(f) printf("Upper Triangle");
11    else printf("Not Upper Triangle");
12 }
```

INPUT

```
1 2 3
0 4 5
0 0 6
```



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QUESTIONS 25 / 25 completed

Low Tri. Not

LOW TRI. NOT

Write a C program to check if a matrix is lower triangular or not.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main(){
3     int a[3][3],i,j,f=1;
4     for(i=0;i<3;i++)
5         for(j=0;j<3;j++)
6             scanf("%d",&a[i][j]);
7     for(i=0;i<3;i++)
8         for(j=i+1;j<3;j++)
9             if(a[i][j]!=0) f=0;
10    if(f) printf("Lower Triangle");
11    else printf("Not Lower Triangle");
12 }
```

INPUT

```
1 0 0
2 3 0
4 5 6
```



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QUESTIONS 25 / 25 completed
Equal
EQUAL
Write a C program to check if two matrices are equal or not.

PROGRAM EDITOR

C

Run

Save

```

1 #include<stdio.h>
2 int main(){
3     int a[3][3],b[3][3],i,j,f=1;
4     for(i=0;i<3;i++)
5         for(j=0;j<3;j++) scanf("%d",&a[i][j]);
6     for(i=0;i<3;i++)
7         for(j=0;j<3;j++) scanf("%d",&b[i][j]);
8     for(i=0;i<3;i++)
9         for(j=0;j<3;j++)
10            if(a[i][j]!=b[i][j]) f=0;
11     if(f) printf("Equal");
12     else printf("Not Equal");
13 }

```

INPUT

```

1 2 3
4 5 6
7 8 9

1 2 3
4 5 6
7 8 6

```



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QUESTIONS 25 / 25 completed
Subtract
SUBTRACT
Write a C program to subtract two matrices.

PROGRAM EDITOR

C

Run

Save

```

1 #include<stdio.h>
2 int main()
3 {
4     int a[3][3]={{3,6,7},{4,6,9},{2,5,8}};
5     int b[3][3]={{2,5,-9},{5,4,6},{8,6,7}};
6     int c[3][3],i,j;
7     for(i=0;i<3;i++)
8     {
9         for(j=0;j<3;j++)
10            {
11                c[i][j]=a[i][j]-b;
12            }
13     }
14     return 0;
15 }

```

INPUT

Your INPUT goes here!



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QUESTIONS 25 / 25 completed
Multiply Scalar
MULTIPLY SCALAR
Write a C program to multiply a matrix by a scalar.

PROGRAM EDITOR

C

Run

Save

```

1 #include<stdio.h>
2 int main(){
3     int a[3][3],i,j,k;
4     for(i=0;i<3;i++)
5         for(j=0;j<3;j++)
6             scanf("%d",&a[i][j]);
7     scanf("%d",&k);
8     for(i=0;i<3;i++){
9         for(j=0;j<3;j++)
10            printf("%d ",a[i][j]*k);
11        printf("\n");
12    }
13    return 0;
14 }

```

INPUT

```

1 2 3
4 5 6
7 8 9
3

```



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QUESTIONS 25 / 25 completed

Determinant

DETERMINANT

Write a C program to find the determinant of a 2x2 matrix.

PROGRAM EDITOR

C

Run Save

```
1 #include<stdio.h>
2 int main(){
3     int a[2][2],d,i,j;
4     for(i=0;i<2;i++)
5         for(j=0;j<2;j++)
6             scanf("%d",&a[i][j]);
7     d=a[0][0]*a[1][1]-a[0][1]*a[1][0];
8     printf("%d",d);
9     return 0;
10 }
```

INPUT

1 2
4 5



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QUESTIONS 25 / 25 completed

Inverse

INVERSE

Write a C program to find the inverse of a 2x2 matrix.

PROGRAM EDITOR

C

Run Save

```
1 #include<stdio.h>
2
3 int main()
4 {
5     int a[2][2];
6     float det;
7     scanf("%d%d%d%d",&a[0][0],&a[0][1],&a[1][0],&a[1][1]);
8     det = a[0][0]*a[1][1] - a[0][1]*a[1][0];
9     if(det == 0)
10     {
11         printf("Inverse does not exist");
12     }
13     else
14     {
15         printf("%.2f %.2f\n", a[1][1]/det, -a[0][1]/det);
16         printf("%.2f %.2f", -a[1][0]/det, a[0][0]/det);
17     }
18     return 0;
19 }
```

INPUT

1 2
3 4



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QUESTIONS 25 / 25 completed

Rank

RANK

Write a C program to find the rank of a matrix.

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[2][2];
6     int det, rank;
7     scanf("%d%d%d%d",&a[0][0],&a[0][1],&a[1][0],&a[1][1]);
8     det = a[0][0]*a[1][1] - a[0][1]*a[1][0];
9     if(det != 0)
10         rank = 2;
11     else if(a[0][0]!=0 || a[0][1]!=0 || a[1][0]!=0 || a[1][1]!=0)
12         rank = 1;
13     else
14         rank = 0;
15     printf("%d", rank);
16     return 0;
17 }
```

INPUT

1 2
3 4

QUESTIONS 25 / 25 completed

Gauss-Jordan

GAUSS-JORDAN

Write a C program to perform Gauss-Jordan elimination on a matrix.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main()
4 {
5     float a[2][3], x, y;
6     for(int i=0; i<2; i++)
7         for(int j=0; j<3; j++)
8             scanf("%f", &a[i][j]);
9     y = (a[1][2] - a[1][0]*a[0][2]/a[0][0]) /
10         (a[1][1] - a[1][0]*a[0][1]/a[0][0]);
11     x = (a[0][2] - a[0][1]*y)/a[0][0];
12     printf("x = %.2f\n", x);
13     printf("y = %.2f", y);
14     return 0;
15 }
```

INPUT

```
2 1 5
1 -1 1
```